

Inhibitory effect of tea polyphenols on local tissue damage induced by snake venoms.



The methanolic extract of fresh tea leaves of *Camellia sinensis* L. (Theaceae) (CS) was assayed for its potential to inhibit enzymes with hydrolytic activity in *Naja naja kaouthia* Lesson (Elapidae) and *Calloselasma rhodostoma* Kuhl (Viperidae) venoms. These snake venom enzymes are responsible for the early effects of envenomation, such as local tissue damage and inflammation. The CS extract inhibited phospholipase A(2), proteases, hyaluronidase and L-amino acid oxidase in both venoms by in vitro neutralization and inhibited the hemorrhagic and the dermonecrotic activities of the venoms in vivo. It is suggested that the inhibitory potential of the CS extract against local tissue damage induced by snake venoms may be attributed to complexation and chelation between the venom proteins and the phenolic contents of the extract.